

From Wholesale Prices to Consumer Inflation: Layered Pass-Through of Global Oil Shocks in India

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Abstract

This paper studies how global oil-price shocks pass through India's domestic price system. Instead of estimating only the effect of oil prices on headline CPI, it follows the shock across wholesale, retail fuel, fuel-sensitive consumer, and headline consumer price layers. Monthly log-difference ADL models are estimated with Newey-West HAC inference and bootstrap symmetry checks. The results show a layered attenuation pattern. Retail petrol and WPI Fuel and Power respond strongly to oil shocks, CPI Fuel and Light gives smaller bridge evidence, and headline WPI shows modest pass-through. At the headline CPI endpoint, positive pass-through is weak and statistically insignificant. The evidence does not support strong short-run asymmetry, except for a marginal retail petrol result. Overall, oil shocks remain visible in fuel layers but lose force in broad consumer inflation.

Keywords: Oil-price pass-through; inflation; India; wholesale price index; consumer price index; retail fuel prices.

JEL codes: C22; E31; Q43; F31.

1. Introduction

India imports a large share of its crude oil and prices that crude in United States dollars. Domestic oil pressure therefore depends on the Brent price and on the rupee-dollar exchange rate, so a useful measure of the domestic oil shock is the rupee-denominated oil price, $\text{oil}^{\text{INR}}_t = \text{Brent}^{\text{USD}}_t \times \text{INR}_t/\text{USD}_t$. When Brent rises or the rupee weakens, the rupee oil price rises, and the cost pressure should appear first in the layers closest to fuel and then, with attenuation, in broader price aggregates (Mandal et al., 2012; Bhanumurthy et al., 2012).

There is a familiar puzzle in this transmission. Petrol and diesel prices visibly track crude, and wholesale fuel inflation moves with rupee oil prices, but headline consumer inflation does not move one-for-one with oil. The Wholesale Price Index (WPI) is closer to producer cost pressure, while the Consumer Price Index (CPI) is the household-facing inflation index and is weighted heavily towards food, housing, services, health, and education. Fuel is only one component of the CPI basket, so even a strong oil shock can be diluted by the time it reaches headline CPI inflation.

Most existing studies on Indian oil pass-through estimate a single equation linking oil to either WPI or CPI inflation (Mandal et al., 2012; Bhanumurthy et al., 2012; Pradeep, 2022). That approach gives one elasticity, but it does not show where the signal weakens. The contribution of this paper is to treat oil-price pass-through as a layered price-system problem rather than a single elasticity. The chain studied here moves from the rupee oil price to wholesale prices, from Brent to the Petroleum Planning and Analysis Cell (PPAC) Delhi retail petrol price, from PPAC retail petrol to CPI Fuel and Light, and from rupee oil to headline CPI. Each layer is estimated as a short-run asymmetric autoregressive distributed lag (ADL) model in monthly log differences.

The institutional background also matters. Domestic fuel prices were not always fully linked to international prices, and reform episodes changed the way international oil movements entered petrol and diesel prices. Pradeep (2022) studies this reform setting and places asymmetry at the centre of the question. This paper uses that background more narrowly. It does not claim that deregulation caused the post-2010 estimates. It treats the reform period as useful context for reading stronger wholesale and retail fuel pass-through after pricing became more market-linked.

The WPI and CPI distinction is useful for this question because the two indices measure different layers of the price system. WPI is closer to producer and wholesale cost pressure. CPI is the consumer-facing aggregate used for inflation targeting, but it contains many items whose monthly movements need not follow oil prices. A layered design therefore gives a more cautious interpretation than a single headline CPI equation. If oil is strong in fuel layers but weak in headline CPI, the correct conclusion is attenuation, not irrelevance.

Inference uses Newey-West heteroskedasticity and autocorrelation consistent (HAC) standard errors (Newey & West, 1987), and the cumulative pass-through statistics CPT^+ and CPT^- summarise the response to positive and negative shock terms. Bootstrap resampling provides a robustness check on asymmetry. Diagnostic tests cover serial correlation, functional form, and recursive stability, and a model-gate rule decides which equations are claim-bearing. Bai and Perron (2003) structural-break logic and KPSS stationarity checks (Kwiatkowski et al., 1992) support the specification choices.

The estimates suggest a layered attenuation pattern. Retail petrol shows the strongest direct fuel response, with $CPT^+ = 0.3459$ ($p < 0.001$). WPI Fuel and Power shows $CPT^+ = 0.2866$ ($p < 0.001$). CPI Fuel and Light, the consumer fuel layer, shows $CPT^+ = 0.1777$ ($p = 0.0021$). Headline WPI shows a statistically significant but modest $CPT^+ = 0.0301$ ($p = 0.0240$). Headline CPI shows $CPT^+ = 0.0213$ with $p = 0.1220$, which is positive in sign but not statistically significant at conventional levels. A formal Wald test rejects equality of the retail petrol and headline CPI positive pass-through estimates at $F = 14.3499$, $p = 0.0002$. Asymmetry is mostly weak, with only retail petrol showing a marginal $CPT^+ = CPT^-$ rejection at the ten per cent level ($p = 0.0999$).

The remainder of the paper is organised as follows. Section 2 describes the data and variables. Section 3 sets out the methodology. Section 4 reports results across price layers. Section 5 covers robustness, diagnostics, and limitations. Section 6 discusses implications and concludes.

2. Data and variables

The dataset is monthly and combines official Indian price series with international oil and exchange-rate data. Headline WPI and WPI Fuel and Power are from the Office of the Economic Adviser (OEA), chained to the 2011-12 base (Office of the Economic Adviser, 2017). Headline CPI and CPI Fuel and Light are from the Ministry of Statistics and Programme Implementation (MoSPI, 2015). Delhi retail petrol prices are from the Petroleum Planning and Analysis Cell (PPAC, 2024). Brent crude is from the World Bank Pink Sheet (World Bank, 2026), and the rupee-dollar exchange rate is taken from a processed exchange-rate source. All series are at monthly frequency.

The domestic oil shock is constructed as $\text{oil}^{\text{INR}}_t = \text{Brent}^{\text{USD}}_t \times \text{INR}/\text{USD}_t$, where INR/USD_t is the monthly average exchange rate. This rupee oil price captures both the Brent component and the exchange-rate component of cost pressure on Indian importers and refiners. For the retail petrol model, the regressor is the Brent shock directly, because domestic retail fuel prices are administered with reference to international product prices and the rupee component is partially absorbed by tax and pricing rules. For the CPI Fuel and Light bridge model, the regressor is the PPAC petrol shock, because that is the price the fuel-sensitive consumer layer faces.

The samples are deliberately allowed to differ by layer. The headline WPI series gives the longest window, from 1983-05 to 2026-03. WPI Fuel and Power begins later, in 1995-05. The PPAC Delhi petrol and headline CPI equations run from 2004-08 to 2024-12 and therefore give the main 20-year consumer-side window. CPI Fuel and Light begins only in 2011-05, so it is used as a bridge between retail fuel and consumer inflation rather than as a mandatory headline result. This is why the title does not state one common date range for every layer.

$$\Delta x_t = 100 \times (\ln(x_t) - \ln(x_{t-1})) \quad (1)$$

Equation (1) defines the monthly log difference of any series x_t . The factor of 100 means coefficients on Δx_t can be read approximately as percentage responses. The positive and negative shock components are defined as $\Delta x^+_t = \max(\Delta x_t, 0)$ and $\Delta x^-_t = \min(\Delta x_t, 0)$. The negative-shock variable is non-positive by construction, so the cumulative pass-through CPT⁻, which is the sum of coefficients on lagged Δx^- terms, should not be interpreted as the response to a positive shock with the opposite sign.

Table 1. Data, transformations, and model role.

Series	Source	Transformation	Sample	Role
Brent crude price	World Bank Pink Sheet	Monthly log difference	Matched to layer	Global oil shock
INR/USD exchange rate	Processed source	Monthly log difference	Matched to layer	Exchange-rate component
Rupee oil price	Brent $\times$ INR/USD	Monthly log difference	Matched to layer	Domestic oil shock
Headline WPI	OEA, base 2011-12	Monthly log difference	1983-05 to 2026-03	Wholesale endpoint
WPI Fuel and Power	OEA, base 2011-12	Monthly log difference	1995-05 to 2026-03	Wholesale fuel layer
PPAC Delhi retail petrol	PPAC	Monthly log difference	2004-08 to 2024-12	Retail fuel layer
CPI Fuel and Light	MoSPI	Monthly log difference	2011-05 to 2024-12	Consumer fuel bridge
Headline CPI	MoSPI/processed CPI	Monthly log difference	2004-08 to 2024-12	Consumer endpoint

Note: All transformations are multiplied by 100, so coefficients can be read approximately as percentage responses. Samples differ because the official series are available over different periods. Source: Author's compilation from cited official sources.

3. Methodology

Each price layer is modelled with a short-run asymmetric ADL specification in monthly log differences. Let Δy_t denote the monthly log change of the layer-specific dependent variable and Δx_t denote the relevant shock variable. The estimating equation is

$$\Delta y_t = \alpha + \sum_{i=1}^p \varphi_i \Delta y_{t-i} + \sum_{j=0}^q \beta_j^+ \Delta x_{t-j}^+ + \sum_{j=0}^q \beta_j^- \Delta x_{t-j}^- + \gamma' Z_t + \mu_m + \varepsilon_t \quad (2)$$

Here Δy_t is monthly inflation in the relevant price index and Δx_t is the layer-appropriate shock. The autoregressive lag length p and the shock lag length q are selected by the Akaike Information Criterion within an upper bound chosen to keep the working sample sufficiently long. The vector Z_t collects optional controls, including activity controls and policy-period dummies, and μ_m denotes month fixed effects that absorb deterministic seasonality. The innovation term ε_t is allowed to be heteroskedastic and serially correlated.

Variables across layers are matched to the channel each layer represents. The headline WPI model uses rupee oil shocks as Δx_t . The WPI Fuel and Power model also uses rupee oil shocks. The PPAC retail petrol model uses Brent shocks. The CPI Fuel and Light bridge model uses PPAC petrol shocks. The headline CPI model uses rupee oil shocks. This matching keeps each regression interpretable as a pass-through equation for its layer.

Cumulative pass-through is summarised by $CPT^+ = \sum_j \beta_j^+$ and $CPT^- = \sum_j \beta_j^-$. Hypothesis tests are reported for $CPT^+ = 0$, $CPT^- = 0$, and $CPT^+ = CPT^-$. Inference uses Newey-West HAC standard errors with the bandwidth chosen by the Newey-West rule (Newey & West, 1987). A restricted-residual circular block bootstrap with 4,999 replications provides a robustness check on the asymmetry test. Predictive precedence between oil and the dependent variable is examined with Granger-style tests on lagged shock terms; these are reported as predictive precedence rather than structural causality. Structural-break behaviour is examined through Bai and Perron (2003) tests, and stationarity of log-differenced series is verified by KPSS tests (Kwiatkowski et al., 1992).

A model is treated as claim-bearing only if it passes a mandatory diagnostic gate, specifically a Breusch-Godfrey test for residual serial correlation, a HAC-robust RESET test for functional form, and a recursive cumulative sum test for parameter stability. Models that fail one of these checks are reported with an explicit caveat or excluded from the main claims. The framework is short-run by design. There are no NARDL bounds tests, no error-correction terms, and no long-run pass-through claims, because the object of interest is monthly transmission rather than long-run equilibrium adjustment.

4. Results

4.1 Cumulative pass-through across price layers

Table 2 reports the central estimates. The reading is layer by layer. The retail petrol model uses Brent shocks over 2004-08 to 2024-12. The wholesale models use rupee oil shocks over the longest available samples. The CPI Fuel and Light bridge uses PPAC petrol shocks over 2011-05 to 2024-12. The headline CPI model uses rupee oil shocks over 2004-08 to 2024-12. Sample sizes vary across layers because the official series are available over different periods.

Table 2. Cumulative pass-through by price layer.

Layer	Sample	N	CPT ⁺	p	CPT ⁻	p	Asym. p	Verdict
PPAC retail petrol	2004-08 to 2024-12	245	0.3459	<0.001	0.1912	0.0002	0.0999	Strong direct fuel; marginal asymmetry
WPI Fuel and Power	1995-05 to 2026-03	371	0.2866	<0.001	0.2677	<0.001	0.7832	Strong wholesale fuel; RESET caveat
CPI Fuel and Light	2011-05 to 2024-12	164	0.1777	0.0021	0.1058	0.1741	0.4554	Bridge evidence; shorter sample
Headline WPI	1983-05 to 2026-03	515	0.0301	0.0240	0.0374	0.0012	0.6727	Modest but significant
Headline CPI	2004-08 to 2024-12	245	0.0213	0.1220	0.0006	0.9375	0.2408	Weak; not significant

Note: CPT⁺ and CPT⁻ are sums of positive- and negative-shock coefficients across the ADL lag window. P-values use Newey-West HAC inference. Asym. p is the p-value for the test CPT⁺ = CPT⁻. WPI models use rupee oil shocks; the retail petrol model uses Brent shocks; the CPI Fuel and Light bridge uses PPAC petrol shocks. Source: Author's estimates from model outputs.

4.2 Wholesale layer

Headline WPI shows statistically significant but modest pass-through from rupee oil shocks. The cumulative response to positive shocks is $CPT^+ = 0.0301$ with $p = 0.0240$, and the cumulative response to negative shock terms is $CPT^- = 0.0374$ with $p = 0.0012$. The asymmetry test does not reject equality ($p = 0.6727$), so the wholesale endpoint does not support short-run asymmetry. The adjusted R^2 of 0.4207 over 515 observations indicates that the lagged inflation, oil shock, and seasonal structure together explain a substantial share of monthly headline WPI variation. The model passes the Breusch-Godfrey, HAC-RESET, and recursive CUSUM checks, so it is treated as claim-bearing.

WPI Fuel and Power carries a much stronger oil signal, with $CPT^+ = 0.2866$ ($p < 0.001$) and $CPT^- = 0.2677$ ($p < 0.001$) over 371 observations. The point estimates imply that approximately 28 to 29 per cent of a one per cent rupee oil shock passes into the wholesale fuel layer in the months captured by the lag window. The asymmetry p-value of 0.7832 is high, so positive and negative effects are statistically indistinguishable. The Breusch-Godfrey and recursive CUSUM tests pass, but the HAC-RESET test for functional form fails. The result is therefore reported as mechanism evidence with a functional-form caveat, rather than as a fully validated structural model.

Figure 1 compares the wholesale pass-through estimates visually. The difference between headline WPI and WPI Fuel and Power is large. This is not surprising, but it is important for interpretation. It means that a broad wholesale aggregate can show only a small average response even when the fuel component inside the same price system is strongly exposed to oil movements. The figure also helps separate economic size from statistical significance. Headline WPI is statistically significant, but the coefficient is small. WPI Fuel and Power is both statistically significant and economically larger, although it carries the RESET caveat noted above.

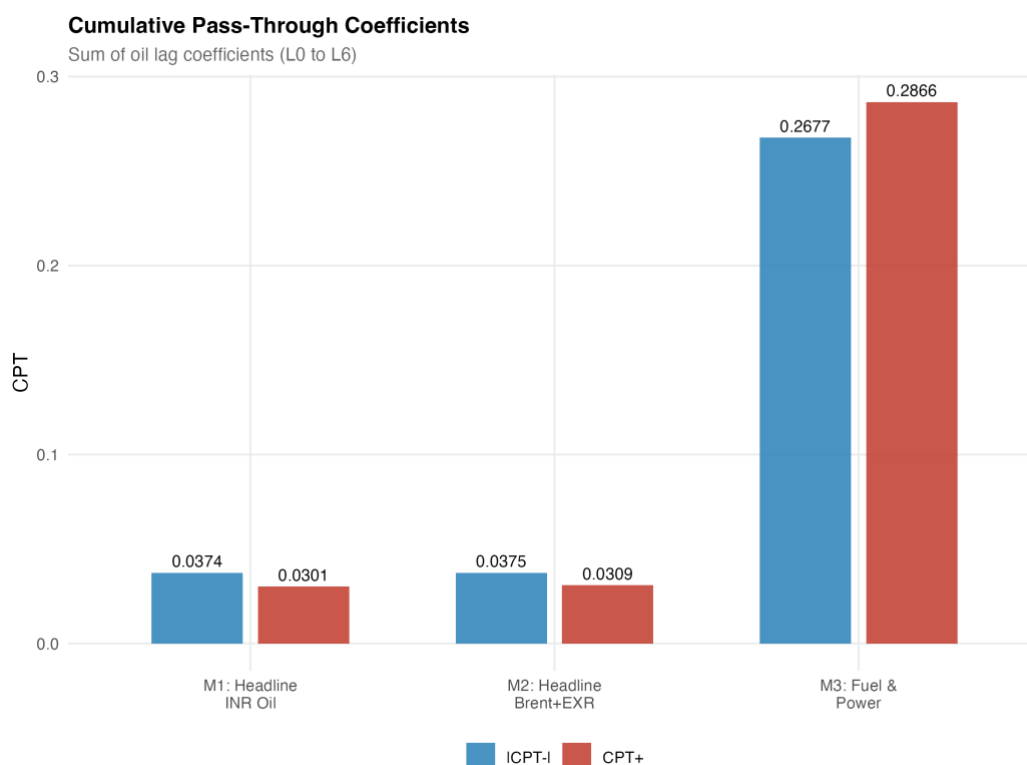


Figure 1. Cumulative oil pass-through in headline WPI and WPI Fuel and Power. Source: Author's estimates from WPI model outputs.

4.3 Retail petrol and the consumer fuel bridge

The PPAC Delhi retail petrol equation is the strongest direct fuel layer in the system. With Brent as the shock variable, $CPT^+ = 0.3459$ ($p < 0.001$) and $CPT^- = 0.1912$ ($p = 0.0002$) over 245 observations. The asymmetry p-value is 0.0999, which rejects equality only at the ten per cent level. The result should therefore be read as suggestive of stronger upward than downward pass-through in retail petrol, but not as decisive evidence of asymmetric pricing. The retail petrol model passes the mandatory mechanism gate, so it can carry the channel claim.

CPI Fuel and Light is the consumer fuel layer that sits between retail fuel prices and broad consumer inflation. With the PPAC petrol shock as the regressor, $CPT^+ = 0.1777$ ($p = 0.0021$) and $CPT^- = 0.1058$ ($p = 0.1741$) over 164 observations. The positive pass-through is statistically significant; the negative side is not. The result is treated as bridge evidence rather than as a primary consumer-price result, because the sample only begins in 2011-05 and is therefore considerably shorter than the headline CPI sample. The estimate is consistent with retail fuel pressure entering the fuel-sensitive consumer layer with attenuation.

The contrast between these two consumer-side layers is useful. Retail petrol is a narrow, policy-sensitive retail price and can move sharply when international prices are passed on. CPI Fuel and Light is already a basket component. It includes household fuel items and therefore smooths the retail petrol shock. The decline from 0.3459 in the retail petrol equation to 0.1777 in the bridge equation is consistent with partial transmission rather than a full one-for-one movement. Because the bridge sample is short, the estimate is not used to make a 20-year headline claim.

4.4 Headline CPI endpoint and integrated attenuation

Headline CPI is where the oil signal becomes weak. The positive cumulative pass-through is $CPT^+ = 0.0213$ with $p = 0.1220$, and the negative side is $CPT^- = 0.0006$ with $p = 0.9375$. The positive coefficient has the expected sign, but it is not statistically significant at conventional levels. Asymmetry is not rejected ($p = 0.2408$). The model passes the mandatory M1 diagnostic gate, so the failure to find significant pass-through is an empirical result, not an artefact of a rejected specification.

The ranking of positive pass-through estimates gives the central result. Retail petrol has the largest CPT^+ , followed by WPI Fuel and Power and CPI Fuel and Light. The broad headline indices are much smaller, especially headline CPI. The formal Wald test for equality of retail petrol and headline CPI positive pass-through rejects the null at $F = 14.3499$, $p = 0.0002$. The evidence is therefore best read as attenuation across price layers rather than absence of oil-price transmission. Figure 2 visualises the layered consumer chain.

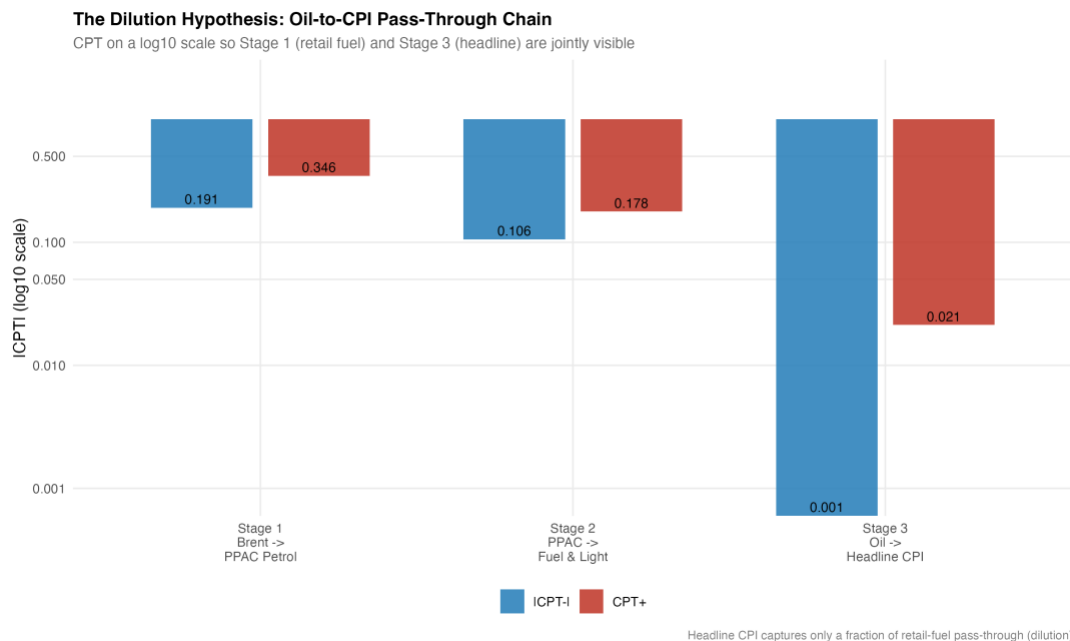


Figure 2. Cumulative positive pass-through across the consumer-price chain. Source: Author's estimates from PPAC and CPI model outputs.

A common-sample exercise re-estimates the consumer chain on a single overlapping window so that sample composition cannot drive the ranking. Stage 1 (Brent to PPAC petrol) gives $CPT^+ = 0.4007$ ($p = 0.0001$). Stage 2 (PPAC petrol to CPI Fuel and Light) gives $CPT^+ = 0.1777$ ($p = 0.0021$). Stage 3 (rupee oil to headline CPI) gives $CPT^+ = 0.0064$ ($p = 0.6355$). The qualitative pattern survives the common-sample restriction.

Predictive-precedence tests give supporting evidence, but they are not used as structural causality tests. In the WPI system, lagged rupee oil changes predict headline WPI inflation ($F = 7.4288$, $p < 0.001$) and WPI Fuel and Power inflation ($F = 15.2231$, $p < 0.001$), while the reverse WPI-to-oil direction is not supported. On the consumer side, Brent strongly predicts PPAC petrol ($F = 10.7127$, $p < 0.001$). Lagged rupee oil terms predict headline CPI only at the ten per cent level ($F = 2.2861$, $p = 0.0794$), and PPAC petrol does not clearly predict CPI Fuel and Light in the bridge equation ($p = 0.1126$). This pattern matches the main estimates: the oil signal is easier to detect near fuel prices than in broad CPI.

5. Robustness, diagnostics, and limitations

The headline WPI result is robust to decomposing the rupee oil shock into a Brent component and an exchange-rate component. The decomposition gives $CPT^+ = 0.0309$ ($p = 0.0234$) and $CPT^- = 0.0375$ ($p < 0.001$), with an asymmetry p -value of 0.7039. These figures are within rounding distance of the rupee-shock benchmark and confirm that the modest wholesale endpoint result does not depend on combining the two components. Bootstrap symmetry tests with 4,999 replications fail to reject $CPT^+ = CPT^-$ for headline WPI and for WPI Fuel and Power, which is consistent with the HAC-based asymmetry tests in Table 2.

The diagnostics are summarised in Table 3. The headline WPI model passes the Breusch-Godfrey test for serial correlation, the HAC-RESET test for functional form, and the recursive CUSUM test for parameter stability. The WPI Fuel and Power model passes Breusch-Godfrey and recursive CUSUM but fails HAC-RESET. The headline CPI M1 model passes the mandatory gate. The PPAC retail petrol model passes the mechanism gate. Two alternative CPI specifications explored in the supporting work, M2 and M3, are not claim-bearing because their diagnostics are rejected. The bootstrap symmetry checks do not reject symmetry in the main wholesale and consumer endpoint models, so asymmetry is not the central finding of the paper.

Table 3. Diagnostics and robustness summary.

Check	Result	Interpretation
Headline WPI diagnostics	BG, HAC-RESET, Rec-CUSUM pass	Main WPI model accepted
WPI Fuel and Power diagnostics	BG, Rec-CUSUM pass; HAC-RESET fails	Mechanism with caveat
Headline CPI M1 gate	Pass	Main CPI endpoint accepted
PPAC mechanism gate	Pass	Direct fuel channel accepted
CPI M2, M3 gates	Diagnostics fail	Excluded from main claims
Bootstrap symmetry (WPI, CPI)	Symmetry not rejected	Asymmetry not central

Note: BG = Breusch-Godfrey; HAC-RESET = HAC-robust Ramsey RESET; Rec-CUSUM = recursive cumulative sum stability test. Source: Author's diagnostic outputs.

A pre/post-2010 split of the wholesale models offers a fourth piece of context. For headline WPI, CPT^+ rises from 0.0117 ($p = 0.3812$) in the pre-2010 sample to 0.0741 ($p = 0.0043$) in the post-2010 sample. For WPI Fuel and Power, CPT^+ rises from 0.0922 ($p = 0.1679$) to 0.5241 ($p < 0.001$). The post-2010 estimates are consistent with stronger pass-through under more market-linked fuel pricing after the petrol deregulation of June 2010 and the diesel deregulation of October 2014. The split is not a clean causal estimate of deregulation, because other macroeconomic and policy changes occurred over the same window. It is reported as institutional evidence.

The limitations are stated plainly. The estimates are reduced-form short-run projections, not structural causal effects. Different layers have different sample spans, although the common-sample exercise reproduces the qualitative ranking. CPI Fuel and Light has a shorter sample than headline CPI and WPI, and is therefore used as bridge rather than primary evidence. PPAC retail petrol uses Delhi prices, which are treated as a retail fuel-channel

proxy rather than a complete national retail measure. The pre/post-2010 split is suggestive institutional evidence, not a clean deregulation experiment. The layered map is not a single mechanical WPI-to-CPI causal chain; it is a set of layer-by-layer pass-through equations that share a common interpretation.

6. Discussion and conclusion

The result of this paper is not that oil is unimportant for Indian inflation. Retail petrol tracks Brent strongly, the wholesale fuel layer responds firmly to rupee oil shocks, and CPI Fuel and Light shows a smaller but statistically significant bridge response. The result is that headline CPI absorbs and dilutes oil shocks because the CPI basket is broad. Food, housing, services, health, and education together dominate the index, and the direct fuel weight, even augmented by indirect cost effects, is too small for the oil signal to dominate the headline figure on a monthly basis.

From a policy perspective, the implication is narrow but useful. A reader who monitors only headline CPI to gauge oil-shock pressure can be misled, because the oil signal is small at that layer and is statistically insignificant in the present sample. Monitoring a small set of fuel-sensitive layers, such as retail petrol, WPI Fuel and Power, and CPI Fuel and Light, carries more information about the speed and size of pass-through than the headline aggregate. This is consistent with earlier work on Indian oil pass-through (Mandal et al., 2012; Bhanumurthy et al., 2012; Pradeep, 2022), and adds the layered organisation as a way to read those results.

The conclusion can be stated compactly. The research question is whether oil-price shocks transmit uniformly across Indian price layers or whether they attenuate as the index becomes broader. The estimates support attenuation: retail petrol > WPI Fuel and Power > CPI Fuel and Light > headline WPI > headline CPI in CPT⁺ terms, with the formal test rejecting equality between the two endpoints of the consumer chain. Asymmetry is not the main finding; only retail petrol shows a marginal asymmetry result at the ten per cent level. The post-2010 split is consistent with stronger pass-through under more market-linked fuel pricing, but it is institutional rather than causal evidence.

The main lesson is that oil shocks do not disappear in India, but they lose force as they move from fuel prices to broad consumer inflation.

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